Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/08/2004
PI Name:	Sutherland, Betsy M Ph.D.		
Project Title:	Complex Space Radiation-Induced DNA Damage Clusters In Human Cell Transformation: Mechanisms, Relationships And Mitigation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bms@bnl.gov	**Fax:**	FY 631/344-3407
PI Organization Type:	GOVERNMENT	**Phone:**	631/344-3380
Organization Name:	Brookhaven National Laboratory		
PI Address 1:	Dept of Biology, Building 463		
PI Address 2:	50 Bell Ave		
PI Web Page:			
City:	Upton	**State:**	NY
Zip Code:	11973-5000	**Congressional District:**	1
Comments:	PI Deceased as of October 2009.		
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	06/01/2003	**End Date:**	05/31/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	5
No. of PhD Candidates:	5	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	NA		
COI Name (Institution):			
Grant/Contract No.:	T-467X		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Predicting and mitigating cancer risk from complex radiation fields in outer space is essential for manned exploration of Mars. This requires understanding the molecular mechanisms by which such radiation inflicts critical damages, responses of cells to radiation, and potentially cancer-inducing changes. Clustered DNA damages are composed of several close-spaced DNA alterations. Specific clusters have different biological impacts, depending on the ability of cells to repair them, and in their cancer-inducing potential. Their levels and types depend on the radiation type, and on factors such as the environment of the DNA. However, repair of clusters in cells exposed to simulated complex radiation fields, the effect of such radiation on induction of potentially oncogenic cellular alterations, and the optimum approaches for mitigating their effects are not known. The project focuses on four experimental series: 1. Determine if prior exposure of human cells to protons affects repair of HZE particle-induced double strand breaks, abasic-, oxidized purine- and oxidized pyrimidine clusters; 2. Test whether dual proton-HZE particle exposures affect neoplastic		

	transformation levels of human cells relative to exposure to each radiation species singly; 3. Determine whether levels of slowly-repaired or unrepaired clusters correlate with human cell neoplastic transformation frequencies; and 4. Test whether factors that reduce cluster induction or increase their repair also decrease levels of neoplastic transformation of human cells. Understanding and mitigating space radiation-induced damage to human cells requires understanding the mechanisms of DNA damage induction and cellular repair, and delineating potential relationships between DNA damage clusters and their cancer-incuding potential. These studies should identify approaches for mitigating potentially cancer inducing effects of space radiation exposure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the role of complex DNA damage clusters in killing and mutating cells can lead to better radiation therapy of cancers. Further, discovering approaches for decreasing DNA damages can minimize the harmful effects of radiation on normal cells.
Task Progress:	During this grant year, the following was achieved: 1. The survival curves for colony formation for human primary cells exposed to radiation doses up to 1 Gy of C (293.8 MeV/n), Fe (1 GeV/n) or Ti (1 GeV/n) were determined. 2. The yields of neoplastic transformation of human primary cells by HZE particles and by protons were determined for Fe, Ti or C particles in producing human cell transformation; duplicate independent experiments were then carried out to further validate these data. 3. The dose-response relations for induction of transformation by protons was determined, using ~ 200 MeV/n protons from Loma Linda University or the Northeast Proton Therapy center at Massachussets General Hospital. 4. Initial dual-radiation exposures were carried out, using cells irradiated with protons as in 3. (above), then 4 days later with Si particles (600 MeV/n). The data indicate that the prior proton exposure did not further increase human cell transformation by Si ions.
Bibliography Type:	Description: (Last Updated: 04/12/2011)
Abstracts for Journals and Proceedings	Sutherland BM, Bennett PV, Zhou G. "Complex space radiation-induced DNA damage clusters in human cell transformation." 15th Annual NASA Space Radiation Health Investigators' Meeting, Port Jefferson, NY, May 2004. 15th Annual NASA Space Radiation Health Investigators' Meeting, Port Jefferson, NY, May 2004. Abstract 97. , May-2004
Abstracts for Journals and Proceedings	Sutherland BM, Bennett PV, Hada M, Kumar P. "DNA damage clusters in low level radiation responses of human cells." 15th Annual NASA Space Radiation Health Investigators' Meeting, Port Jefferson, NY, May 2004. 15th Annual NASA Space Radiation Health Investigators' Meeting, Port Jefferson, NY, May 2004. Abstract 45. , May-2004
Presentation	Sutherland, B. "Clustered DNA Damages." US Department of Energy review of research. Feb-2004